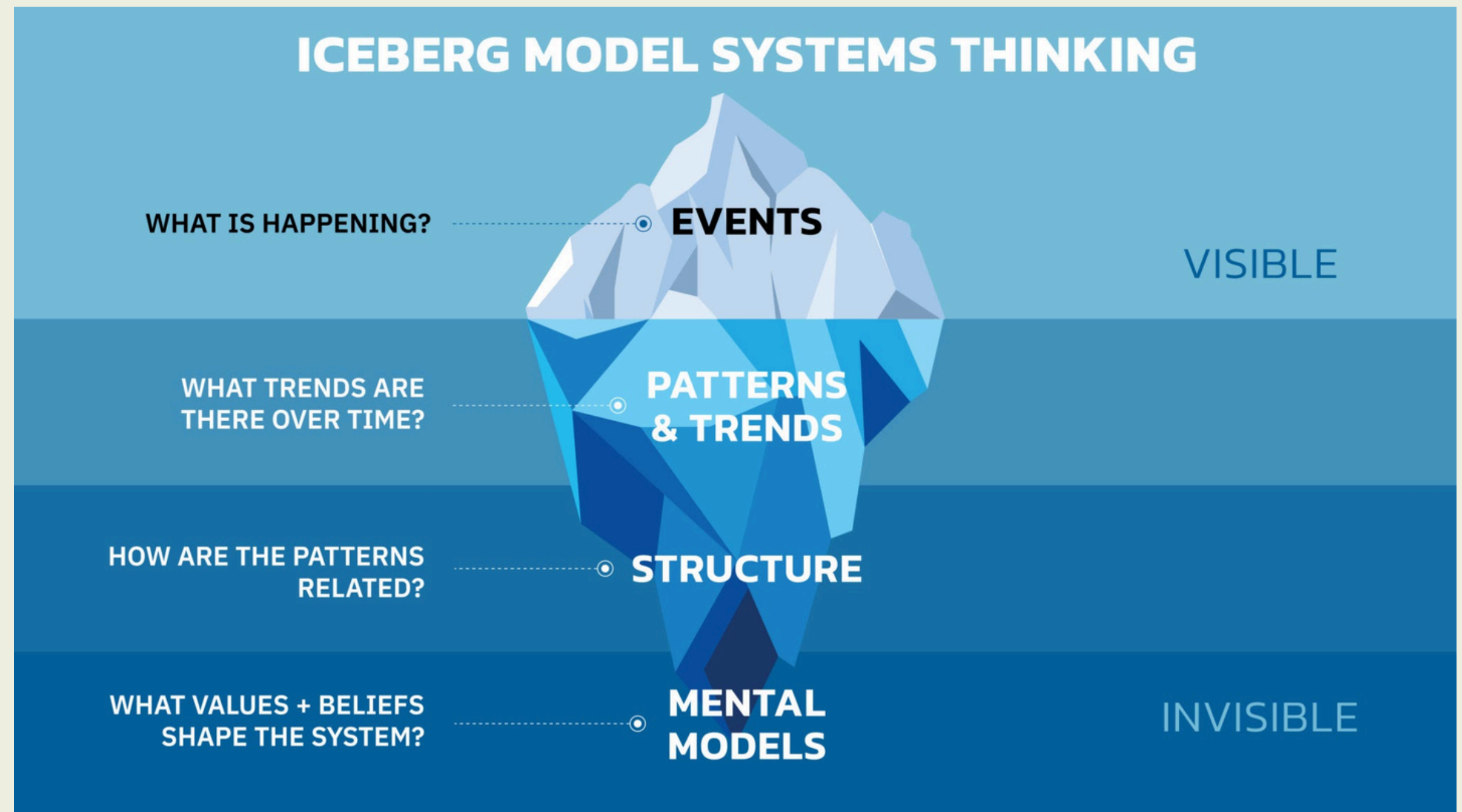


LECTURE 4

The Iceberg Model (The "Why")

The Iceberg Model (The "Why")

Most people or organisation – live entirely above the waterline, reacting to events one at a time. Systems thinkers learn to dive, look deeper at *Patterns, Systemic Structures, and finally, Mental Models.*



The Iceberg Model (The "Why")

“When something goes wrong, where do we usually intervene?”

They will naturally arrive at:

Events

What happened?

Patterns

What keeps happening?

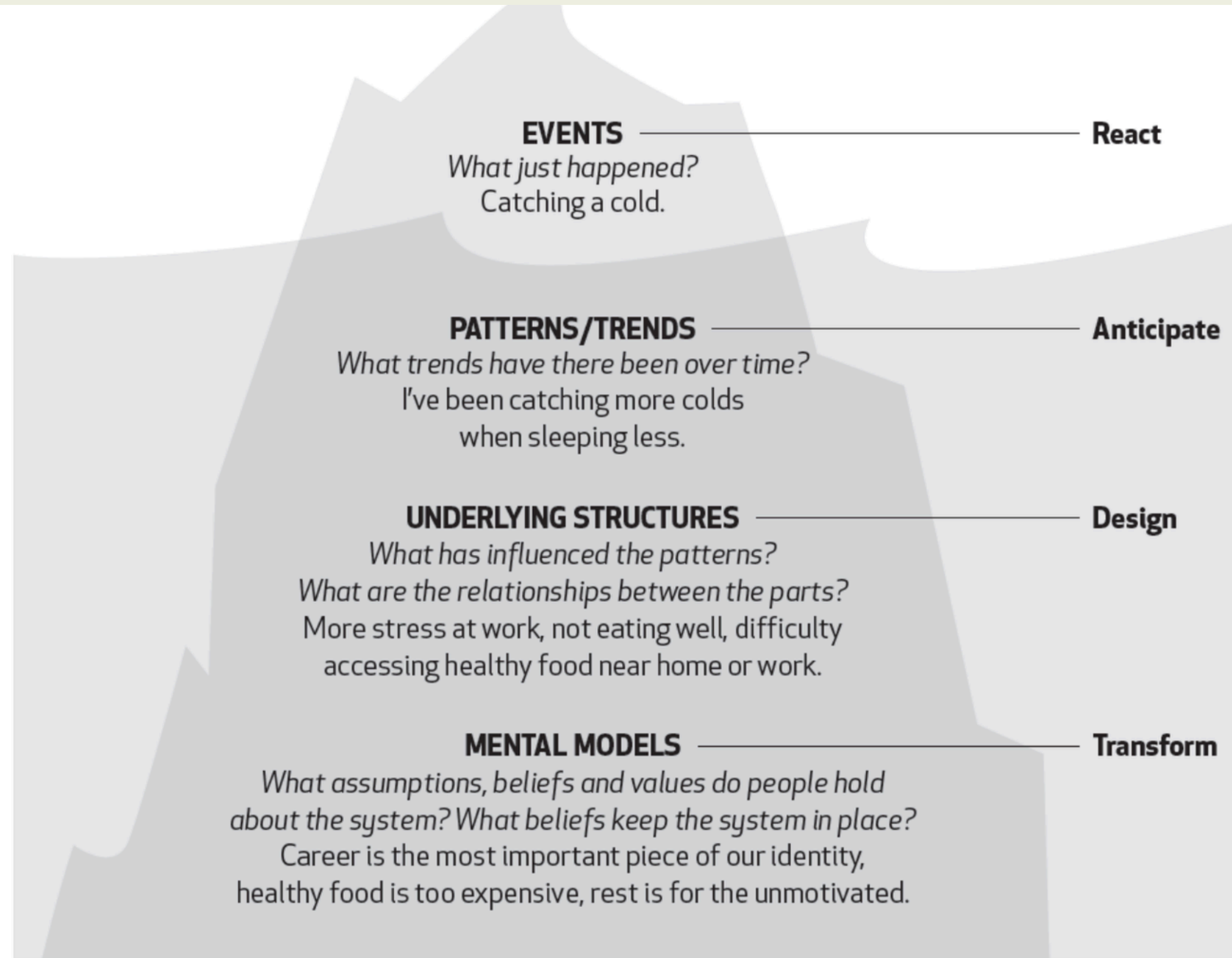
Systemic Structures

What is causing this pattern?

Mental Models

What beliefs/assumptions created the structure?

This is precisely the distinction emphasised by the Donella Meadows Project: the visible event is only the tip; patterns, structures and mental models lie beneath it and help explain why events recur.



The Iceberg Model (The "Why")

“Students keep submitting assignments late.”

Event:

Students submitted assignments late.

Pattern:

The same students—or many students—are repeatedly late.

Systemic Structure:

Multiple deadlines, unclear workload coordination, grading incentives, lack of feedback, timetable conflicts.

Mental Model:

“Students perform only when deadlines and penalties force them to.”

The Iceberg Model (The "Why")

An event asks us to fix what happened.

Systems thinking asks us to understand what keeps making it happen.

00:00 – 00:10 **OPENING HOOK**

SYSTEM THINKING AND DESIGN

Every day,
the same traffic jam.
We react. We wait. We get late.
But why does it keep happening?

00:10 – 00:20 **ABOVE THE WATERLINE – EVENTS**

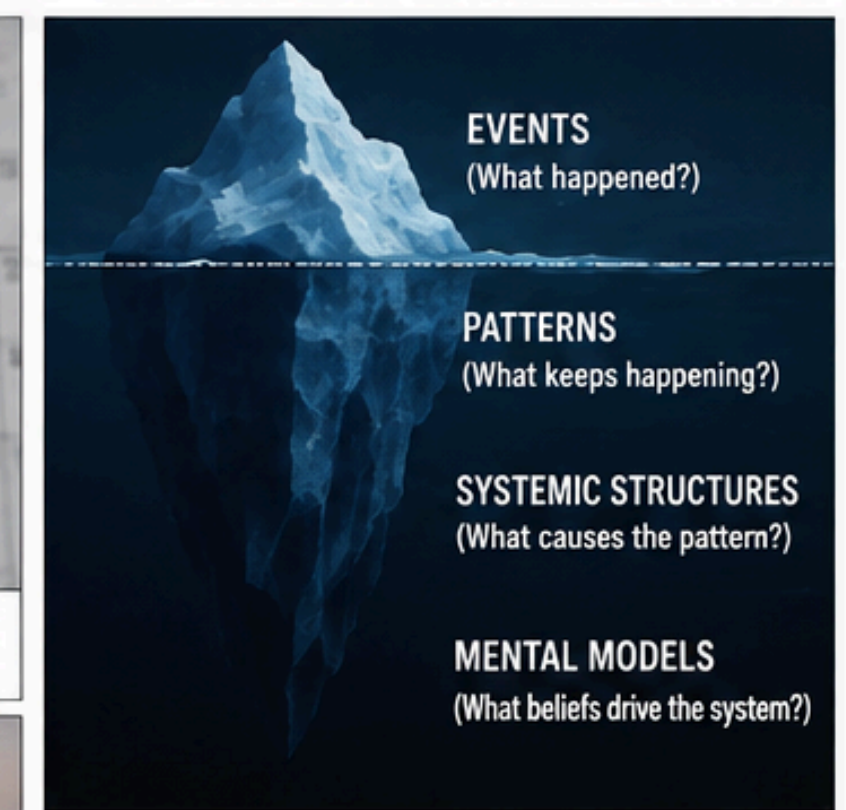
EVENTS
What happened?
There is a traffic jam.

We see the jam. We feel the problem.

00:20 – 00:40 **GO DEEPER – PATTERNS**

PATTERNS
What keeps happening?
Jams happen every day,
at the same places,
at the same times.

It's not a one-time event.
It's a pattern.



00:40 – 01:10 **DEEPER – SYSTEMIC STRUCTURES**

SYSTEMIC STRUCTURES
What causes the pattern?

- More vehicles, limited road space
- Poor public transport
- Last-mile connectivity issues
- Uncoordinated traffic signals
- No demand management
- Urban planning focused on cars

The pattern is created by the way
our system is designed.

01:10 – 01:40 **DEEPEST – MENTAL MODELS**

MENTAL MODELS
What beliefs drive
the system?

- My time is more important
- Owning a car = success
- Roads are for vehicles,
not for people
- Short-term thinking
- Responsibility is
someone else's job

These beliefs shape the structures.
And the structures create the jams.

01:40 – 02:05 **SO WHAT?**

If we only react to events,
we get temporary relief.

If we work on structures,
we see change.

If we shift our mental models,
we transform the system.

Systems thinking helps us
solve the right problem.

02:05 – 02:25

REACTIVE
Blame. Complain. Wait.
Do the same.
Get the same result.

DESIGNER
Understand. Redesign. Collaborate.
Create better choices.
Get better results.

We can choose to be victims of the system,
or designers of a better one.

02:25 – 02:35

Next time you face a problem,
dive deeper.
Look below the surface.
That's where real solutions live.

Dive. Discover. Design. Transform.

02:35 – 02:45

EVENTS

PATTERNS

SYSTEMIC STRUCTURES

MENTAL MODELS

See the system.
Understand the system.
Change the system.
Transform the future.

Systems thinking. Better questions. Better world.

How to use in Class (Discussion Prompts)

- What are the events (above the waterline)?
- What patterns do you notice?
- What structures are causing the pattern?
- What mental models are driving the structures?
- Where can we intervene for lasting change?

Key Takeaway

Most people live above the waterline
and react to events.

Systems thinkers dive deep and
work on what really matters.